INTEGRAL ELEMENTS FROM TWO EQUATIONS: ČECH COHOMOLOGY AND GCDS, CONSTRUCTIVELY

Abstract. Let k be a discrete field, $A = k[a, b]$ and $I = (a, b)$. For two equations $p = x+af+bg$, $q = y+au+bv$ in $A[x, y]$ we construct, in every A-algebra where p and q vanish, an element $s \equiv 1 \pmod I$ such that s, sx and sy are integral over A. The proof is constructive. It computes in an explicit Čech model of the projective closure of $p = q = 0$, lifts a fixed cochain by a finite correction whose existence follows from the constructive noetherianity of A, and reads the integral equations on global sections. Two finiteness statements replace the coherence of higher direct images: one uses a single element of the elimination ideal, the other uses the gcd of the homogenised equations. For the latter we prove that $R[X]$ is a gcd domain when R is one, using the direct content induction of Mines–Richman–Ruitenburg, and that a divisor of a homogeneous polynomial is homogeneous.

1. Statement

All rings are commutative. A *discrete field* is a ring with $1 \neq 0$ in which every element is 0 or invertible, and this alternative is decidable.

Theorem 1.1. *Let k be a discrete field, $A = k[a, b]$ and $I = (a, b)$. Let*

$$P = \mu_P + aF_P + bG_P, \qquad Q = \mu_Q + aF_Q + bG_Q$$

be homogeneous polynomials in $A[X, Y, Z]$ of degrees $d, e \geq 1$, where $\mu_P = XZ^{d-1}$ and $\mu_Q = YZ^{e-1}$, and let $p = P(x, y, 1)$, $q = Q(x, y, 1)$. Let L be an A-algebra and $x, y \in L$ with $p(x, y) = q(x, y) = 0$. Then there are $r, t, s \in L$ with

$$s = 1 + ar + bt$$

such that s, sx and sy are roots of monic polynomials in $A[T]$.

Taking $L = A[x, y]/(p, q)$ gives the statement for the algebra defined by the two equations. The existence is not uniform: s is obtained after a well-founded search, through the constructive noetherianity of A. No regularity of the pair P, Q is assumed; they may have a common factor (for instance $p = x(1 + ax)$, $q = y(1 + ax)$, where $Z + aX$ divides P and Q).

Plan. Section 2 proves that $R[X]$ is a gcd domain when R is one, so that $S = k[a, b][X, Y, Z]$ is a gcd domain. Section 3 transports the gcd of P, Q to Laurent polynomials, with homogeneous factors. Section 4 describes the Čech model. Section 5 lifts the cochain Z^{-1} to a section of $\mathcal{O}_W(-1)$, and Section 6 produces the integral equations. Section 7 follows the construction on an example.

2. Gcd domains

Definition 2.1. A ring R is a *discrete domain* if one can decide $x = 0$ for every x, $1 \neq 0$, and $x \neq 0$, $y \neq 0$ imply $xy \neq 0$. For $c, x \in R$, a witness of $c \mid x$ is some q with $x = cq$; divisibility is never decided. A *gcd* of x, y is an element d with $d \mid x$, $d \mid y$ and $c \mid d$ for every common divisor c of x, y. A *gcd domain* is a discrete domain together with a function giving a gcd of any two elements.

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This is the notion of [2]. A gcd is unique up to a unit only; it is data, not a property. Note $\gcd(0,0)=0$. A discrete field is a gcd domain: $\gcd(x,y)=y$ if $x=0$ and x otherwise.

Lemma 2.2. *In a gcd domain, $d \mid ux$ and $d \mid uy$ imply $d \mid u\gcd(x,y)$.*

Proof. If $u=0$ there is nothing to prove. Otherwise let $g=\gcd(ux,uy)$, so $d \mid g$, and $u \mid g$, say $g=ut$. From $ut \mid ux$ we get $t \mid x$ by cancelling u, and likewise $t \mid y$; hence $t \mid \gcd(x,y)$ and $g=ut \mid u\gcd(x,y)$. □

Corollary 2.3. *If $\gcd(x,y) \mid 1$ and $x \mid yz$, then $x \mid z$.*

Proof. $x \mid zx$ and $x \mid zy$, so $x \mid z\gcd(x,y) \mid z$. □

Content. For a sequence $e_0, e_1, \dots$ put $\mathrm{c}_0(e)=0$ and $\mathrm{c}_{n+1}(e)=\gcd(e_n, \mathrm{c}_n(e))$. By induction, $\mathrm{c}_n(e)$ divides e_i for $i<n$, every common divisor of these e_i divides $\mathrm{c}_n(e)$, and, by Lemma 2.2, $d \mid ue_i$ for all $i<n$ gives $d \mid u\mathrm{c}_n(e)$. For $f \in R[X]$ of degree $<n$ let $\mathrm{c}(f)=\mathrm{c}_n(f_0,f_1,\dots)$. Call f *primitive* if every common divisor of its coefficients divides 1.

Lemma 2.4. *(i) If f is primitive and $d \mid uf_i$ for all i, then $d \mid u$. (ii) If $f \neq 0$, then $\mathrm{c}(f) \neq 0$ and $f=\mathrm{c}(f)f^*$ with f^* primitive.*

Proof. (i) $d \mid u\mathrm{c}(f)$ and $\mathrm{c}(f) \mid 1$. (ii) If $\mathrm{c}(f)=0$ all coefficients vanish. Divide each coefficient by $\mathrm{c}(f)$. If d divides the coefficients of f^*, then $d\mathrm{c}(f)$ divides those of f, hence $\mathrm{c}(f)$, and $d \mid 1$ after cancelling $\mathrm{c}(f)$. □

Lemma 2.5 (Gauss)**.** *If $E \in R[X]$ is primitive and d divides every coefficient of Eq, then d divides every coefficient of q.*

Proof. We use the direct induction of [2, IV, §4, Lemma 4.3]. Our coefficient lists are constant-first, so we remove the constant term rather than the leading term. Induct on the sum of the list lengths to prove the stronger statement

$$d \mid \text{every coefficient of } fg \quad\Longrightarrow\quad d \mid \mathrm{c}(f)\mathrm{c}(g). \tag{C}$$

First suppose $f=a+Xf_1$ and $g=b+Xg_1$ are primitive, and c divides all coefficients of fg. Put $t=\gcd(c,a)$. Subtracting ag from fg shows that t divides every coefficient of f_1g. The induction hypothesis gives $t \mid \mathrm{c}(f_1)\mathrm{c}(g)$, hence $t \mid \mathrm{c}(f_1)$ since g is primitive. Together with $t \mid a$, this says that t divides all coefficients of f, so $t \mid 1$. Symmetrically $\gcd(c,b) \mid 1$. As $c \mid ab$, the constant coefficient of the product, Corollary 2.3 gives $c \mid b$. Thus $c \mid \gcd(c,b) \mid 1$: the product is primitive.

For general f,g, if either content is zero, the conclusion of (C) is immediate. Otherwise write $f=uf^*$ and $g=vg^*$ with $u=\mathrm{c}(f)$, $v=\mathrm{c}(g)$ and f^*,g^* primitive. Scalar division preserves the lengths of the lists, so the primitive-product argument just proved applies to f^*,g^* at this same induction step. Now $fg=uv(f^*g^*)$, and Lemma 2.4(i), applied to the primitive product, gives $d \mid uv$. Empty lists are handled directly; they cannot be primitive since $1 \neq 0$.

Apply (C) to E,q. Since $\mathrm{c}(E) \mid 1$, we get $d \mid \mathrm{c}(q)$, and hence $d \mid q_j$ for every j. □

The same argument proves content multiplicativity up to associates:

$$\mathrm{c}(fg) \sim \mathrm{c}(f)\mathrm{c}(g).$$

Statement (C) gives one divisibility; the other follows by multiplying common coefficient divisors. No prime element, factorisation or Dedekind–Mertens lemma is used, and gcds need not be normalised.

Lemma 2.6. *If E is primitive, $\lambda \neq 0$ and $\lambda f=Eq$, then $E \mid f$.*

Proof. By Lemma 2.5, λ divides the coefficients of q: $q = \lambda q'$. Then $\lambda(f - Eq') = 0$, so $f = Eq'$. □

Lemma 2.7 (elimination)**.** *For $f, g \in R[X]$ one can construct $D \in R[X]$ and $\lambda \neq 0$ in R with $D \mid \lambda f$, $D \mid \lambda g$, and such that every common divisor of f and g divides D.*

Proof. Induction on $n_f + n_g$, where f (resp. g) has degree $< n_f$ (resp. $< n_g$). If $n_f = 0$, then $f = 0$: take $D = g$ and $\lambda = 1$; the case $n_g = 0$ is symmetric. If $f_{n_f-1} = 0$, decrease n_f; likewise for g. Otherwise $\alpha = f_{n_f-1}$ and $\beta = g_{n_g-1}$ are non-zero. Suppose $n_f \geq n_g$ and put $k = n_f - n_g$ and $f' = \beta f - \alpha X^k g$, of degree $< n_f - 1$. Induction gives D and λ' for f', g. Take $\lambda = \lambda'\beta \neq 0$. Then $\lambda f = \lambda' f' + \alpha X^k \lambda' g$ and $\lambda g = \beta \lambda' g$ are divisible by D, and a common divisor of f, g divides f', hence D. If $n_f < n_g$, exchange the roles of f and g. □

The field of fractions of R is not used.

Theorem 2.8. *If R is a gcd domain, so is $R[X]$.*

Proof. Discrete domain. $f = 0$ is decidable, since f has finitely many coefficients. Suppose $f \neq 0$, $g \neq 0$ and $fg = 0$. Statement (C), with $d = 0$, gives $\mathrm{c}(f)\mathrm{c}(g) = 0$. Both contents are nonzero by Lemma 2.4(ii), contradicting the domain property of R.

Gcd. If $f = 0$ take g, and if $g = 0$ take f. Otherwise let D, λ be as in Lemma 2.7. Then $D \neq 0$, so $D = \mathrm{c}(D)E$ with E primitive, and $E \mid f$, $E \mid g$ by Lemma 2.6. Put $\gamma = \gcd(\mathrm{c}(f), \mathrm{c}(g))$ and $G = \gamma E$.

G divides f: write $f = Eq$. Then γ divides the coefficients of f, hence those of q by Lemma 2.5. So $q = \gamma q'$ and $f = Gq'$. The same holds for g.

Let h divide f and g. Then $h \neq 0$ and $h = \mathrm{c}(h)h^*$. The coefficients of $f = hr$ are divisible by $\mathrm{c}(h)$, so $\mathrm{c}(h) \mid \mathrm{c}(f)$; likewise $\mathrm{c}(h) \mid \mathrm{c}(g)$, hence $\mathrm{c}(h) \mid \gamma$. Moreover $h \mid D = \mathrm{c}(D)E$, so $h^* \mid \mathrm{c}(D)E$, and $h^* \mid E$ by Lemma 2.6. Therefore $h = \mathrm{c}(h)h^* \mid \gamma E = G$. □

Corollary 2.9. *For a discrete field k, the ring $S = k[a, b, X, Y, Z]$ is a gcd domain.*

In a gcd domain, let $P \neq 0$ and Q be given, $D_0 = \gcd(P, Q)$, $P = D_0 P'$ and $Q = D_0 Q'$. Then $D_0 \neq 0$ and $\gcd(P', Q') \mid 1$: a common divisor c of P', Q' gives $D_0 c \mid D_0$. Hence

(S0) $D_0 f = 0$ implies $f = 0$;
(S1) $P'\alpha + Q'\beta = 0$ implies $\beta = P'w$ and $\alpha = -Q'w$ for a unique w.

Indeed $P' \mid Q'\beta$ gives $\beta = P'w$ (Corollary 2.3); then $P'(\alpha + Q'w) = 0$, and $P' \neq 0$.

3. Homogeneous factors and Laurent polynomials

Let $\Lambda = A[X^{\pm 1}, Y^{\pm 1}, Z^{\pm 1}]$. We represent its elements by their coefficient functions $\mathbf{Z}^3 \to A$ with finite support; the degree of $m \in \mathbf{Z}^3$ is $m_X + m_Y + m_Z$, and f is homogeneous of degree k if $f(m) = 0$ whenever $\deg m \neq k$. Let $\psi : S \to \Lambda$ send a polynomial to its coefficient function, extended by 0 outside $\mathbf{N}^3$. Then ψ is A-linear, injective and multiplicative, and multiplication by X, Y, Z corresponds to the shifts of exponents.

Lemma 3.1 (clearing denominators)**.** *For every $f \in \Lambda$ there is $K \geq 0$ such that $(XYZ)^K f = \psi(s)$ for some $s \in S$: take K at least the sum of the negative parts of the exponents in a finite list representing f.*

Lemma 3.2. *If $s \in S$ is non-zero, then multiplication by $\psi(s)$ is injective on Λ.*

Proof. If $\psi(s)f = 0$, write $(XYZ)^K f = \psi(s')$. Then $\psi(ss') = 0$, so $ss' = 0$, so $s' = 0$ since S is a domain, and therefore $f = 0$. □

For $x \in S$ let x_n be its homogeneous component of degree n. Since S is discrete, $x_n = 0$ is decidable, and every $x \neq 0$ has a *top degree* T and a *bottom degree* $B \leq T$: $x_T \neq 0$, $x_B \neq 0$, and $x_n = 0$ for $n > T$ and for $n < B$. These are found by a finite search below a bound on the degrees of the terms of x.

Proposition 3.3. *Let $x, y \in S$ be non-zero with xy homogeneous of degree d. Then x and y are homogeneous.*

Proof. Let T, T' be the top degrees of x, y, $h = x_T$ and $h' = y_{T'}$. Then $xy = hh' + h(y - h') + (x - h)y$. The last two terms have no component of degree $\geq T + T'$, so the component of xy of degree $T + T'$ is hh', which is non-zero since S is a domain. Hence $T + T' = d$. In the same way the bottom degrees satisfy $B + B' = d$. Since $B \leq T$ and $B' \leq T'$, we get $B = T$ and $B' = T'$, so x and y are homogeneous. □

Apply this to $P = D_0 P'$ and $Q = D_0 Q'$, with P, Q as in Theorem 1.1 (they are non-zero, since the coefficient of μ_P in P is 1). Then D_0, P', Q' are homogeneous, of degrees r, $d' = d - r$ and $e' = e - r$. Through ψ we obtain the following statements in Λ:

Proposition 3.4. (S0) *If $D_0 f = 0$ in Λ, then $f = 0$.*

(S1) *If $\alpha, \beta \in \Lambda$ are homogeneous of degrees $k + e'$ and $k + d'$ with $P'\alpha + Q'\beta = 0$, then there is a unique $w \in \Lambda$, homogeneous of degree k, with $\alpha = -Q'w$ and $\beta = P'w$.*

Proof. (S0) is Lemma 3.2. For (S1), multiply α, β by the same $(XYZ)^K$ to land in $\psi(S)$, apply (S1) of Section 2 in S, and divide back by $(XYZ)^K$. Uniqueness follows from Lemma 3.2 for P'. Since P' and Q' are homogeneous, the component of w of degree k is again a solution, hence equal to w. □

4. The Čech model

For a non-empty set E of variables, let $\Lambda_E(k)$ be the elements of Λ of degree k whose exponents of the variables outside E are ≥ 0. Put

$$K^0(k) = \Lambda_X(k) \times \Lambda_Y(k) \times \Lambda_Z(k), \quad K^1(k) = \Lambda_{XY}(k) \times \Lambda_{XZ}(k) \times \Lambda_{YZ}(k), \quad K^2(k) = \Lambda_{XYZ}(k),$$

with the usual Čech differentials δ_0, δ_1. Two j-cochains are *congruent* if their difference is $Pu + Qv$, componentwise, with u, v cochains satisfying the same support conditions; we write $N^j(k)$ for these differences. Then N^j is a submodule stable under δ and under multiplication by X, Y, Z. A 0-cochain c is *closed* if $\delta_0 c \in N^1$. The closed cochains of degree 0 form a ring under the componentwise product, with unit $(1, 1, 1)$, and $N^0(0)$ is an ideal of this ring.

We use three explicit operators, each acting coefficient by coefficient: the projection π of an element of Λ_{XYZ} onto its all-negative monomials, a section $\sigma : K^2 \to K^1$ of δ_1, and a contraction $\tau : K^1 \to K^0$. They satisfy, in every degree,

$$q = \pi(q) + \delta_1 \sigma(q), \qquad \delta_0 \tau(b) = b \ \text{ if } \delta_1 b = 0$$

(the second identity says $H^1(\mathcal{O}(k)) = 0$). Moreover $\pi(\delta_1 b) = 0$ for every 1-cochain b. Only exponents are compared in these constructions, never coefficients.

Restriction to the chart Z. For $f \in \Lambda_Z(k)$ let $\mathrm{res}(f) = \sum c\, x^{m_X} y^{m_Y} \in L$ over the terms cX^m of f. It is additive and multiplicative, and $\mathrm{res}(P) = p(x, y) = 0$, $\mathrm{res}(Q) = 0$. Hence the map $c \mapsto \mathrm{res}(c_Z)$ is a ring map from closed sections of degree 0 to L which kills $N^0(0)$. It satisfies $\mathrm{res}(Xw) = x\,\mathrm{res}(Zw)$ and $\mathrm{res}(Yw) = y\,\mathrm{res}(Zw)$ for $w \in K^0(-1)$.

5. Lifting the cochain Z^{-1}

Let $z_* = (0, 0, Z^{-1}) \in K^0(-1)$. Then $\delta_0 z_* = (0, Z^{-1}, Z^{-1})$.

Lemma 5.1 (fixed cochain)**.** *For every n there are $w_n \in I^n K^1(-1)$ and an explicit congruence $\delta_0 z_* \equiv w_n$.*

Proof. On the chart XZ put $\theta = \mu_P^{-1}$ and $E = a(-\theta F_P) + b(-\theta G_P) = 1 - \theta P$. Then $Z^{-1} = E^n Z^{-1} + P u_n$ with $u_n = \theta \sum_{i<n} E^i Z^{-1}$. Similarly, on YZ, with $\theta' = \mu_Q^{-1}$ and $E' = 1 - \theta' Q$. Take $w_n = (0, E^n Z^{-1}, E'^n Z^{-1})$, which lies in $I^n K^1(-1)$ through the expansion of E^n and E'^n. □

Lemma 5.2 (one element of the elimination ideal)**.** *Let $g = \mu_Q P - \mu_P Q$ and* gens *the finite set of all-negative exponents t of degree $-\deg g - 1$. Put $m_t = \sigma(gX^t)$. For every $n \geq 1$ there are $\rho_n \in A^{\mathrm{gens}}$ with $\rho_{n,t} \in I^{n-1}$, and $v_n \in I^n K^0(-1)$, such that*

$$\delta_0 z_* \equiv \sum_t \rho_{n,t} m_t + \delta_0 v_n.$$

Proof. $\delta_1 w_n = (E'^n - E^n) Z^{-1}$ and $E' - E = \theta P - \theta' Q = g/(\mu_P \mu_Q)$. Hence $\delta_1 w_n = g q_n$ with $q_n = (\mu_P \mu_Q)^{-1} \sum_{i<n} E'^i E^{n-1-i} Z^{-1}$, of degree $-\deg g - 1$, whose coefficients lie in I^{n-1}. Since $\pi(g q_n) = \pi(\delta_1 w_n) = 0$, the decomposition $q = \pi(q) + \delta_1 \sigma(q)$ applied to gX^t and $\delta_0 \tau = \mathrm{id}$ on cocycles give

$$w_n = g\, \sigma(q_n) + \sum_t q_n(t)\, \sigma(gX^t) + \delta_0 c_n$$

with $c_n = \tau(\cdots) \in I^n K^0(-1)$. The first term is congruent to 0 since $g \in (P, Q)$. Put $\rho_{n,t} = q_n(t)$ and $v_n = c_n$, and use Lemma 5.1. □

Note that $\rho_{n,t} \in I^{n-1}$, so the class of $\delta_0 z_*$ in degree n of the Rees module $\bigoplus_n H^1(I^n \mathcal{O}_W(-1))$ is generated by the m_t in Rees degree 1. No Gröbner basis and no other element of the elimination ideal is needed.

Lemma 5.3 (finite correction)**.** *Let $N \geq 1$, $r_i \in I^{N-i}$ for $1 \leq i < N$, and $v \in I^N K^0(-1)$ with $\delta_0 z_* - \sum_{i<N} r_i \delta_0 z_* \equiv \delta_0 v$. Then $h = (1 - \sum_i r_i) z_* - v$ is closed and $h - z_* \in I K^0(-1)$.*

Proof. $\delta_0 h \equiv 0$ by linearity, and $r_i \in I$, $v \in I K^0$ since $N - i \geq 1$. □

Proposition 5.4. *There is a closed $h \in K^0(-1)$ with $h - z_* \in I K^0(-1)$.*

Proof. $A = k[a, b]$ is noetherian in the constructive sense: every sequence of elements of a finitely generated module, given by codes in a coded presentation, has a term that is a linear combination of the previous ones, and this term is reached by a well-founded search. This follows from the noetherianity of a discrete field, the transfer along $k \to A$, and the Hilbert basis theorem [1]. Apply it to the Rees algebra $\bigoplus_n I^n T^n \subseteq A[T]$, coded inside $A[T] \oplus A[T]^{\mathrm{gens}}$, and to the sequence ρ_n of Lemma 5.2. Taking the component of degree N of the relation found gives N and $r_i \in I^{N-i}$ with $\rho_N = \sum_{i<N} r_i \rho_i$. Then

$$\delta_0 z_* - \sum_{i<N} r_i \delta_0 z_* \equiv \delta_0 \Big(v_N - \sum_{i<N} r_i v_i \Big),$$

with $v = v_N - \sum r_i v_i \in I^N K^0(-1)$. Conclude by Lemma 5.3. □

6. Integral equations

Let $\delta = d + e - r$, where $r = \deg D_0$, and let $\mathcal{G}$ be the all-negative exponents of degree $-\delta$. For $t \in \mathbf{Z}^3$ write X^t for the corresponding monomial, and put

$$c_t = \tau\big(P\,\sigma(-Q'X^t) + Q\,\sigma(P'X^t)\big), \qquad t \in \mathcal{G}.$$

For an element $\omega = \sum_t \omega_t X^t$ of degree $-\delta$ with $\pi(Q'\omega) = \pi(P'\omega) = 0$, the cochain $c_\omega = \sum_t \omega_t c_t$ is closed of degree 0, with $\delta_0 c_\omega = P\sigma(-Q'\omega) + Q\sigma(P'\omega)$ exactly. Indeed δ_1 of the argument of τ is $-PQ'\omega + QP'\omega = 0$, because $PQ' = QP' = D_0P'Q'$.

Proposition 6.1 ((F2))**.** *For every closed $c \in K^0(0)$ one constructs $\kappa(c) = (a_0, (\omega_t)) \in A^{1+\mathcal{G}}$ with $c \equiv a_0 \cdot 1 + \sum_t \omega_t c_t$.*

Proof. Write $\delta_0 c = Pu + Qv$. Then $0 = \delta_1\delta_0 c = D_0(P'\delta_1 u + Q'\delta_1 v)$, so $P'\delta_1 u + Q'\delta_1 v = 0$ by (S0), and (S1) gives w of degree $-\delta$ with $\delta_1 u = -Q'w$ and $\delta_1 v = P'w$. Put $\omega = \pi(w)$ and $\omega_t = w(t)$. Since $w = \omega + \delta_1\sigma(w)$,

$$\delta_1(u + Q'\sigma(w)) = -Q'\omega, \qquad \delta_1(v - P'\sigma(w)) = P'\omega,$$

so $\pi(Q'\omega) = \pi(P'\omega) = 0$. Let $u'' = u + Q'\sigma(w) - \sigma(-Q'\omega)$ and $v'' = v - P'\sigma(w) - \sigma(P'\omega)$. They are cocycles, hence $u'' = \delta_0\tau(u'')$ and $v'' = \delta_0\tau(v'')$. Using $QP' = PQ'$ one finds

$$\delta_0 c = \delta_0\big(c_\omega + P\tau(u'') + Q\tau(v'')\big),$$

so $c' = c - c_\omega - P\tau(u'') - Q\tau(v'')$ satisfies $\delta_0 c' = 0$ exactly. A 0-cocycle of degree 0 satisfying the support conditions is a constant: its three components agree, and every exponent is ≥ 0 with degree 0. Hence $c' = a_0 \cdot 1$, and $c \equiv a_0 + c_\omega$. □

Proof of Theorem 1.1. Let h be given by Proposition 5.4, and let c be one of the closed sections Zh, Xh, Yh of degree 0. The powers c^i are closed. Apply the constructive noetherianity of A to the sequence $\kappa(c^i)$ in $A^{1+\mathcal{G}}$ (formed with the witnesses of closedness of the products). It gives M and $\lambda_i \in A$ with $\kappa(c^M) = \sum_{i<M} \lambda_i \kappa(c^i)$. The map $(a_0, \omega) \mapsto a_0 + c_\omega$ is linear, so

$$c^M \equiv \sum_{i<M} \lambda_i c^i \pmod{N^0}.$$

The restriction is a ring map that kills N^0, so $\operatorname{res}(c)$ is a root of $T^M - \sum_i \lambda_i T^i$. Put $s = \operatorname{res}(Zh)$. Then $\operatorname{res}(Xh) = xs$ and $\operatorname{res}(Yh) = ys$, so s, sx, sy are integral over A. Finally $h = (1 - \sum r_i)z_* - v$ with $r_i \in I$ and $v \in IK^0(-1)$, and $\operatorname{res}(Zz_*) = 1$, so $s = 1 - \sum r_i - \operatorname{res}(Zv) \in 1 + IL$. □

Remark 6.2. The argument uses only the following data about P and Q: the shape $P = \mu_P + aF_P + bG_P$, $Q = \mu_Q + aF_Q + bG_Q$ (for Lemma 5.1), and Proposition 3.4 (for Proposition 6.1). When D_0 is a unit, (S0) is empty and (S1) is the coprimality of P, Q; when D_0 is not a unit, as for $p = x(1 + ax)$, $q = y(1 + ax)$, the quotients P', Q' enter only in the syzygy, while P and Q still define the congruences and the sections c_t.

7. An example

We follow the construction on

$$p = x - a + bxy + 2bx^2, \qquad q = y - b + ax^2 + axy + by^2,$$

so that $P = XZ - aZ^2 + b(XY + 2X^2)$ and $Q = YZ + a(X^2 + XY) + b(Y^2 - Z^2)$, with $d = e = 2$. The numbers below were obtained with an auxiliary computer algebra script, which checks every identity it uses (in particular $\delta_1 w_n = gq_n$, $\delta_1 R = 0$ and $\delta_0\tau(R) = R$ for the cochain R of Lemma 5.2).

The gcd. $\gcd(P,Q)=1$, so (S0) is empty and (S1) is the coprimality of P and Q. Thus $\delta=4$ and $\mathcal{G}$ consists of the three all-negative exponents of degree -4; the closed sections are generated by 1 and the three c_t, in accordance with the Bézout number $de=4$.

The generators of (F1). $g=YZP-XZQ$ has degree 4, so there are six generators m_t, indexed by the all-negative exponents of degree -5, ordered $(-1,-1,-3)$, $(-1,-2,-2)$, $(-1,-3,-1)$, $(-2,-1,-2)$, $(-2,-2,-1)$, $(-3,-1,-1)$. The first coordinates are

$$\begin{aligned}\rho_1&=(1,0,0,0,0,0),\\ \rho_2&=(0,b,0,a,0,0),\\ \rho_3&=(-(a+b)(a+3b),0,b^2,0,ab,a^2),\\ \rho_4&=(0,-a^3-2a^2b-4ab^2-4b^3,0,-a(a^2+6ab+4b^2),0,0).\end{aligned}$$

The odd terms live on four generators and the even terms on the other two.

The relation. Since ρ_n is homogeneous of degree $n-1$ in a,b, a relation $\rho_N=\sum_{i<N} r_i\rho_i$ with $r_i\in I^{N-i}$ is a linear system over $\mathbf{Q}$. There is none for $N\le 5$. For $N=6$ one finds

$$\rho_6=r_2\rho_2+r_4\rho_4,\qquad r_2=-(a^4+6a^3b+10a^2b^2+32ab^3+9b^4)\in I^4,\quad r_4=-2(a+b)(a+3b)\in I^2,$$

and $r_1=r_3=r_5=0$. (The proof does not prescribe which relation is used; a search through the constructive noetherianity of A may stop at another index.)

The element s. The correction is $h=(1-r_2-r_4)z_*-v$ with $v=v_6-r_2v_2-r_4v_4$, and $s=\operatorname{res}(Zh)=1-r_2-r_4-\operatorname{res}(Zv)$ is a polynomial in x,y of total degree 6, with 52 terms and coefficients of degree at most 6 in a,b. Modulo I^4,

$$s\equiv 1+2(a+b)(a+3b)=1-r_4,$$

so $s\equiv 1 \pmod{I^2}$.

Integrality. Over $\mathbf{Q}(a,b)$ the algebra $A[x,y]/(p,q)$ has dimension 4, with basis $1,y,y^2,x$. The characteristic polynomials of s, sx and sy are monic of degree 4 with coefficients in $\mathbf{Q}[a,b]$, as Theorem 1.1 predicts.

Comparison. An element with the same properties is $u=(1+ax+by)(1+2bx+by)^2$. The element s produced by the construction is different: $s-u\notin(p,q)$. Both satisfy equations of degree 4, but that of s is much larger; for instance its constant term has degree 24 in a,b, against 12 for u. The construction is uniform in p,q, but it does not look for a small certificate.

8. Formalisation

Everything above is formalised and checked in the proof checker Unit, in the developments `sag/gcd` (Sections 2 and 3) and `sag/acl-zmt` (Sections 4–6). The foundations are those of the other developments of `sag`: univalence and replacement are parameters, function extensionality is derived, and $A[X]$ is a set quotient of coefficient lists. No other axiom, no choice principle and no excluded middle is used. The main checked statements are:

- Statement (C): `Gauss.Direct.product`, by direct induction. Content multiplicativity is given by the declarations `content-mul-left` and `content-mul-right`. The existing `GaussAt.gauss` and `DivPrim.result` interfaces are retained;
- `PolyGcd.Over[R,G].domain`: Theorem 2.8; `Iterate`: Corollary 2.9;
- `HomDivisor.Divisor`: Proposition 3.3; `HomLambda.Data`: Proposition 3.4, with the homogeneous factors as lists of terms;
- `CechSections.Sec.F2D`: Proposition 6.1; `CechZMTD0`, `ACLGeneral`: the assembly over a gcd domain with a coded noetherian presentation;
- `ACLGeneralK.Over[k,...].Charts[...].certificate`: Theorem 1.1, with the charts μ_P,μ_Q slightly more general than XZ^{d-1}, YZ^{e-1};

- `ACLFinalGeneral` and `ACLControlGeneral`: the instances $p = x - a + bxy + 2bx^2$, $q = y - b + ax^2 + axy + by^2$, and $p = x(1 + ax)$, $q = y(1 + ax)$, where all hypotheses are discharged by computation.

The computations of Section 7 are not part of the formalisation; the script that produced them is `sag/acl-zmt/acl_cech_trace.py`. Divisibility is never decided. The gcd is computed by Lemma 2.7, and the degree of a polynomial, being unique, is extracted from the set quotient. The gcd entry point imports neither `DedekindMertens` nor `ZeroProduct`. Fresh checks of the revised sources passed: 3975 declarations for `sag/gcd` and 7274 for `sag/acl-zmt`, with no failures. The opening audit for the modified gcd proofs is `sag/gcd/gcd-direct.opens`.

References

[1] T. Coquand and H. Persson, *Gröbner bases in type theory*, in Types for Proofs and Programs, LNCS 1657, Springer, 1999.

[2] R. Mines, F. Richman and W. Ruitenburg, *A Course in Constructive Algebra*, Springer, 1988.